Chapter 3: Exponentials & Logarithms

Student Question Booklet

Sections 3F–3L : Year 10 Mathematical Methods : 2026

How to use this booklet

This booklet is your **map through sections 3F–3L** — exponential graphs, growth and decay, compound interest and logarithms. For every section you will find:

- **Worked examples** — fully solved examples with a diagram where it helps, plus the matching textbook example numbers and pages to cross-reference.
- **Question sets** — the questions to practise, grouped by *Building Understanding*, *Fluency*, *Problem-solving*, *Reasoning* and *Enrichment*, with page numbers. Find them in your own textbook.
- **Hurdle questions** — five questions you **must** be able to do to pass the section, with working space (and a grid wherever you need to sketch).

Diagrams are schematic (not to scale). Sections marked 10A are extension. Your calculator's log button is base 10. Round as instructed and always show working — write the rule before you calculate.

Page references are to *Essential Mathematics for the Victorian Curriculum 10 & 10A*, 3rd ed. Question sets belong to the textbook; the worked examples, diagrams and hurdle questions are original material written for this course.

3F Exponential relations and their graphs CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 16** — Sketching graphs of exponentials (p. 251)
- **Example 17** — Sketching with reflections (p. 252)
- **Example 18** — Solving exponential equations graphically (p. 252)

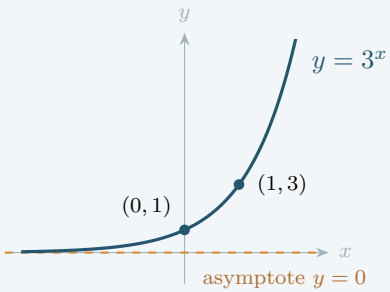
Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 250–251)
- **Fluency:** Q1–4 (p. 253)
- **Problem-solving:** Q5–8 (p. 253)
- **Reasoning:** Q9–12 (p. 254)
- **Enrichment: $y = 2^{-x}$ and $y = \left(\frac{1}{2}\right)^x$:** Q13 (p. 254)

Worked examples (study the method, then try the hurdles)

Sketching an exponential graph

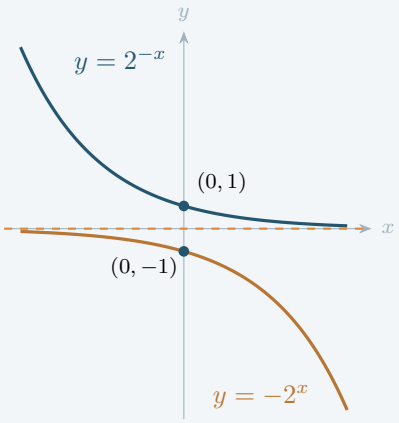
Sketch $y = 3^x$. Label the y-intercept, the point where $x = 1$, and the asymptote.



Solution. Every graph $y = a^x$ (with $a > 1$) passes through $(0, 1)$ because $a^0 = 1$. At $x = 1$, $y = 3^1 = 3$, giving $(1, 3)$. For negative x the values get very small ($3^{-2} = \frac{1}{9}$) but never reach 0, so the x-axis, $y = 0$, is the **asymptote**. The curve rises steeply for $x > 0$.

Reflecting an exponential graph

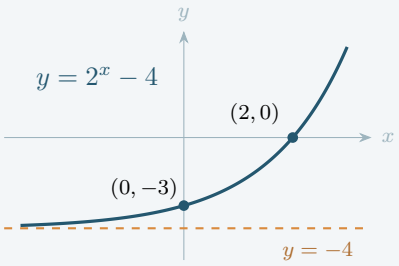
On the same axes, sketch $y = 2^{-x}$ and $y = -2^{x}$. State the reflection that produces each graph from $y = 2^{x}$.



Solution. Replacing x with $-x$ reflects the graph in the y-**axis**: $y = 2^{-x}$ still passes through $(0, 1)$ and has asymptote $y = 0$, but it *decreases*. Putting a negative in front, $y = -2^{x}$, reflects the graph in the x-**axis**: it passes through $(0, -1)$, keeps the asymptote $y = 0$, and lies below it.

A translated exponential graph

For $y = 2^{x} - 4$, find the asymptote, the y-intercept and the x-intercept, then sketch the graph.



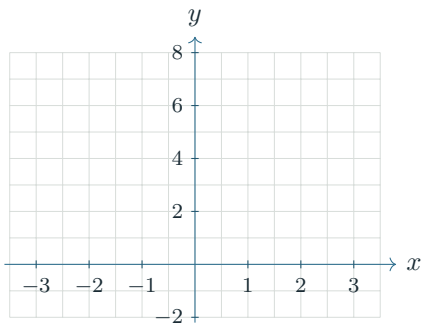
Solution. Subtracting 4 moves the whole graph, *including the asymptote*, down 4: the asymptote is $y = -4$. When $x = 0$, $y = 1 - 4 = -3$, so the y-intercept is $(0, -3)$. When $y = 0$:

$$2^{x} - 4 = 0 \;\Rightarrow\; 2^{x} = 4 = 2^{2} \;\Rightarrow\; x = 2,$$

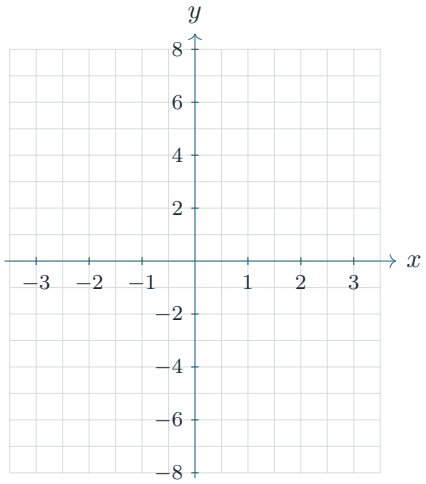
so the x-intercept is $(2, 0)$.

Hurdle questions — Section 3F (show all working)

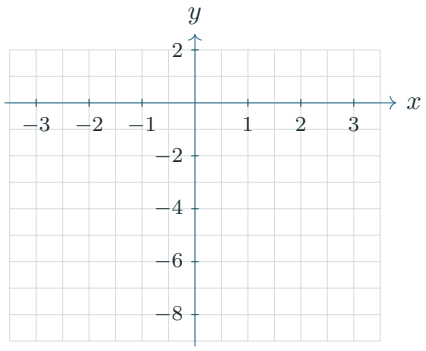
H1. Sketch the graph of $y = 4^x$ on the grid. Label the y-intercept, the point where $x = 1$, and the asymptote.



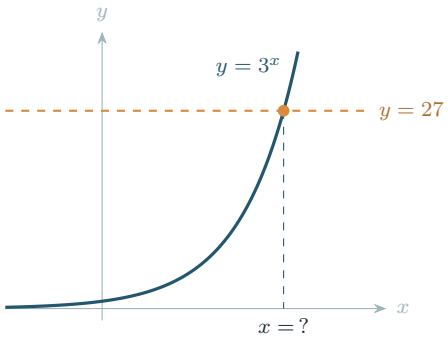
H2. On the same axes, sketch $y = 3^{-x}$ and $y = -3^x$. Label each y-intercept and the asymptote, and state which axis each graph is a reflection of $y = 3^x$ in.



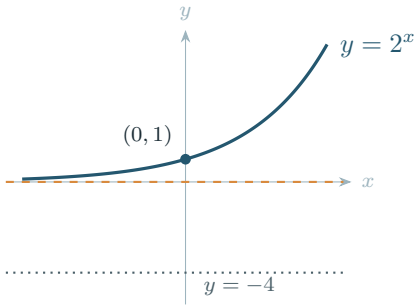
H3. For the graph of $y = 2^x - 8$, find: **(a)** the equation of the asymptote, **(b)** the y-intercept, **(c)** the x-intercept. Then sketch the graph, labelling all three features.



H4. The graphs of $y = 3^x$ and $y = 27$ meet at one point. Find its coordinates *without* a calculator, and explain your method using powers of 3.



H5. Explain why the equation $2^x = -4$ has no solution. Refer to the graph of $y = 2^x$ and its asymptote in your answer.



3G Exponential growth and decay CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 19** — Writing exponential rules (p. 258)
- **Example 20** — Applying exponential rules (p. 258)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 257)
- **Fluency:** Q1–6 (p. 259–260)
- **Problem-solving:** Q7–9 (p. 260–261)
- **Reasoning:** Q10–11 (p. 261)
- **Enrichment: Half-life:** Q12–14 (p. 261)

Worked examples (study the method, then try the hurdles)

Writing growth and decay rules

Write a rule for each situation, defining your variables. **(a)** A colony of 500 bacteria grows by 20% every hour. **(b)** A swimming pool holding 12 000 L loses 5% of its water each day.

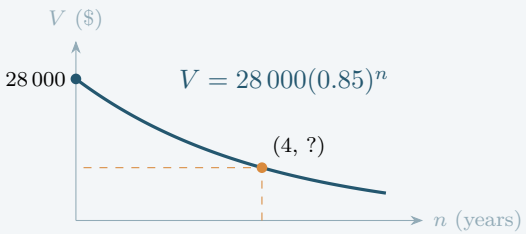
Solution. Growing by $r\%$ means multiplying by $1 + \frac{r}{100}$ each period; losing $r\%$ means multiplying by $1 - \frac{r}{100}$.

(a) N = number of bacteria after t hours: $N = 500(1.2)^t$ (growth, $\times 1.2$)

(b) V = volume in litres after t days: $V = 12\,000(0.95)^t$ (decay, $\times 0.95$)

Applying a decay rule

A car bought for \$28 000 loses 15% of its value each year. Write a rule for its value and find the value after 4 years, to the nearest dollar.



Solution. A 15% loss each year means multiplying by 0.85, so $V = 28\,000(0.85)^n$. Substitute $n = 4$:

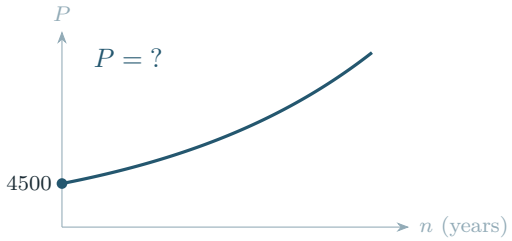
$$V = 28\,000 \times 0.85^4 = 28\,000 \times 0.52200625 = 14\,616.175\ldots$$

After 4 years the car is worth about **\$14 616**.

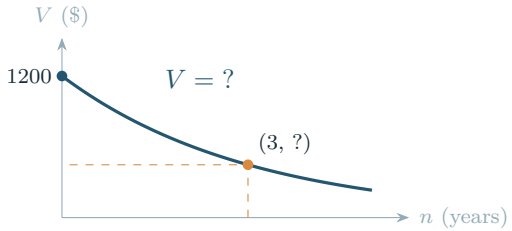
Hurdle questions — Section 3G (show all working)

H1. Write an exponential rule for each situation, defining your variables. **(a)** \$2500 grows by 6% per year. **(b)** An 80 g radioactive sample decays by 9% per hour. **(c)** A population of 12 000 falls by 2.5% per year.

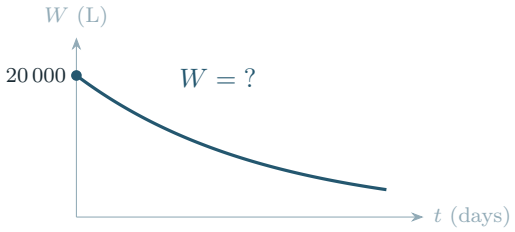
H2. A town has 4500 people and its population grows by 2% per year. **(a)** Write a rule for the population P after n years. **(b)** Find the population after 10 years, to the nearest whole person.



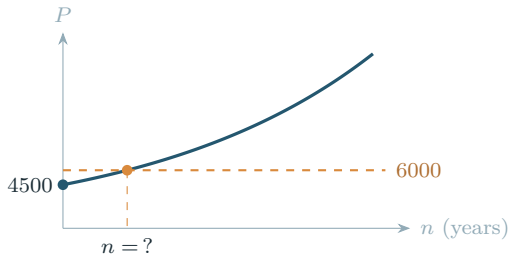
H3. A new phone costs \$1200 and loses 25% of its value each year. Find its value after 3 years.



H4. A water tank holding 20 000 L has a leak, and loses 4% of its water each day. How much water is left after one week? Give your answer to the nearest litre.



H5. For the town in H2, use your rule and a table of values (trial and error) to find the first whole number of years after which the population is more than 6000.



n						
P						

3H Compound interest CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 21** — Using the compound interest formula (p. 264)
- **Example 22** — Converting rates and time periods (p. 264)
- **Example 23** — Finding compounded amounts using months (p. 265)

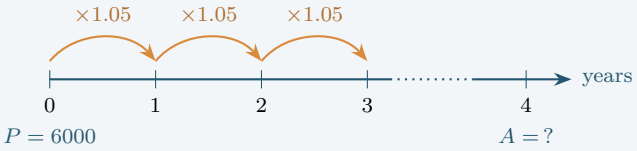
Question sets (practise from your textbook)

- **Building Understanding:** Q1–4 (p. 263)
- **Fluency:** Q1–4 (p. 265–266)
- **Problem-solving:** Q5–7 (p. 266)
- **Reasoning:** Q8–9 (p. 266)
- **Enrichment: Double your money:** Q10 (p. 266)

Worked examples (study the method, then try the hurdles)

Using the compound interest formula

\$6000 is invested at 5% p.a., compounded annually, for 4 years. Find the final amount and the interest earned, to the nearest cent.



Solution. Identify $P = 6000$, $r = 5$, $n = 4$ and use $A = P\left(1 + \frac{r}{100}\right)^n$.

$$A = 6000(1.05)^4 = 6000 \times 1.21550625 = 7293.04,$$
$$\text{Interest} = A - P = 7293.04 - 6000 = 1293.04.$$

The investment grows to **\$7293.04**, earning **\$1293.04** in interest.

Compounding monthly

\$10 000 is invested at 6% p.a., compounded *monthly*, for 3 years. Find the final amount, to the nearest cent.

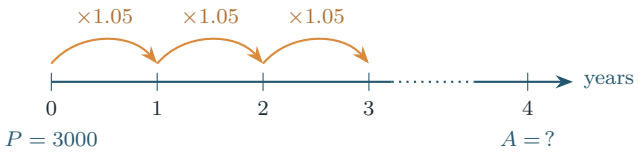
Solution. Convert to monthly periods first: the rate per month is $r = 6 \div 12 = 0.5$ and the number of periods is $n = 3 \times 12 = 36$.

$$A = 10\,000\left(1 + \tfrac{0.5}{100}\right)^{36} = 10\,000(1.005)^{36} = 11\,966.81.$$

The final amount is **\$11 966.81**.

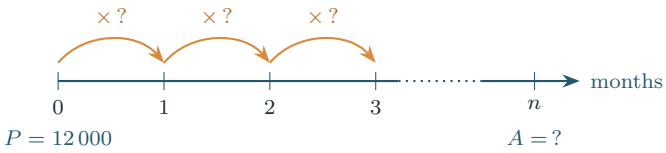
Hurdle questions — Section 3H (show all working)

H1. \$3000 is invested at 5% p.a., compounded annually, for 4 years. Find the final amount and the interest earned, each to the nearest cent.

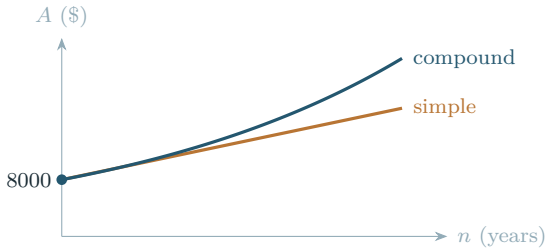


H2. For each investment, state the number of periods n and the interest rate r per period. **(a)** 8% p.a. for 5 years, compounded quarterly. **(b)** 4.8% p.a. for 2 years, compounded monthly.

H3. \$12 000 is invested at 6% p.a., compounded monthly, for 5 years. Find the final amount, to the nearest cent.



H4. \$8000 is invested for 6 years at 5% p.a. Find the final amount if the interest is **(a)** simple interest, and **(b)** compounded annually. **(c)** How much more does compound interest earn? *(Simple interest:* $I = \frac{Prn}{100}$*.)*



H5. Jai has \$10 000 to invest for 3 years. Option A pays 6% p.a. compounded annually; Option B pays 5.8% p.a. compounded monthly. Calculate the final amount for each option and state which is better.

3I Introducing logarithms CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 24** — Writing equivalent statements involving logarithms (p. 268)
- **Example 25** — Evaluating logarithms (p. 268)
- **Example 26** — Solving simple logarithmic equations (p. 269)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 268)
- **Fluency:** Q1–5 (p. 270)
- **Problem-solving:** Q6–8 (p. 270–271)
- **Reasoning:** Q9–11 (p. 271)
- **Enrichment: Fractional logarithms:** Q12 (p. 271)

Worked examples (study the method, then try the hurdles)

Switching between index form and log form

(a) Write $3^4 = 81$ in logarithmic form. **(b)** Write $\log_2 32 = 5$ in index form.

$$\log_3 81 = 4 \quad \xleftrightarrow{\text{same fact}} \quad 3^4 = 81$$

log form — index form

the log is the **power**

Solution. $\log_a x = y$ and $a^y = x$ say the same thing: the base stays the base, and the log is the **power**.

(a) $3^4 = 81 \Leftrightarrow \log_3 81 = 4$ (b) $\log_2 32 = 5 \Leftrightarrow 2^5 = 32$

Evaluating logarithms and solving log equations

Evaluate **(a)** $\log_5 125$ and **(b)** $\log_{10} 0.001$ without a calculator. Then solve **(c)** $\log_3 x = 4$ and **(d)** $\log_x 49 = 2$.

Solution. Read each log as the question "*base to what power gives the number?*" Write equations in index form.

(a) $5^3 = 125$, so $\log_5 125 = 3$.

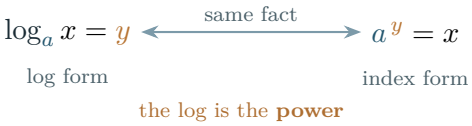
(b) $0.001 = \frac{1}{1000} = 10^{-3}$, so $\log_{10} 0.001 = -3$.

(c) $\log_3 x = 4$ means $x = 3^4$, so $x = 81$.

(d) $\log_x 49 = 2$ means $x^2 = 49$, so $x = 7$ (a base must be positive).

Hurdle questions — Section 3I (show all working)

H1. Write each statement in **logarithmic form**: **(a)** $2^6 = 64$, **(b)** $10^4 = 10\,000$, **(c)** $5^{-2} = \frac{1}{25}$.



H2. Write each statement in **index form**: **(a)** $\log_3 243 = 5$, **(b)** $\log_{10} 0.01 = -2$, **(c)** $\log_6 1 = 0$.

H3. Evaluate without a calculator: **(a)** $\log_2 128$, **(b)** $\log_{10} 100\,000$, **(c)** $\log_4 \frac{1}{16}$.

H4. Solve for x: **(a)** $\log_5 x = 3$, **(b)** $\log_x 64 = 3$, **(c)** $\log_7 x = 0$.

H5. Use your calculator to evaluate $\log_{10} 50$, correct to three decimal places. Then explain, using powers of 10, why the answer must lie between 1 and 2.



3J Logarithmic scales CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 27** — Interpreting a logarithmic chart (p. 276)
- **Example 28** — Constructing a logarithmic graph (p. 278)

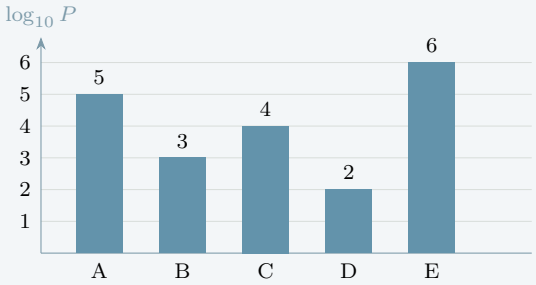
Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 275–276)
- **Fluency:** Q1–5 (p. 279–280)
- **Problem-solving:** Q6–8 (p. 281–282)
- **Reasoning:** Q9–11 (p. 282–283)
- **Enrichment: Exploring pH:** Q12 (p. 283)

Worked examples (study the method, then try the hurdles)

Reading a logarithmic chart

The chart shows $\log_{10} P$, where P is the population of five towns. **(a)** What is the difference in order of magnitude between town A and town D? **(b)** How many times larger is the population of town A than town D?

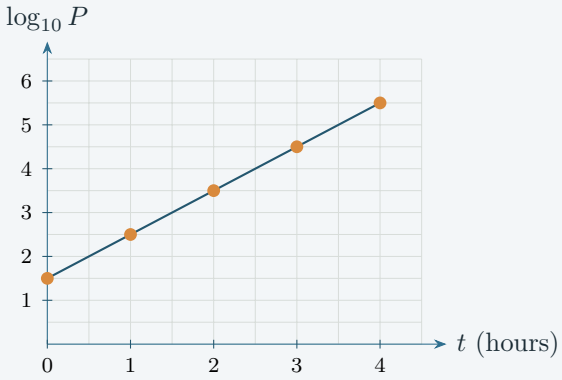


Solution. The height of each bar is the **order of magnitude** (the power of 10). **(a)** $5 - 2 = 3$. **(b)** Each step on a log scale multiplies by 10, so a difference of 3 means $10^3 = 1000$. Town A's population is 1000 **times** larger than town D's.

Constructing a logarithmic graph

A culture of bacteria has P cells after t hours. Find $\log_{10} P$ for each value (one decimal place), plot $\log_{10} P$ against t, and describe the relationship.

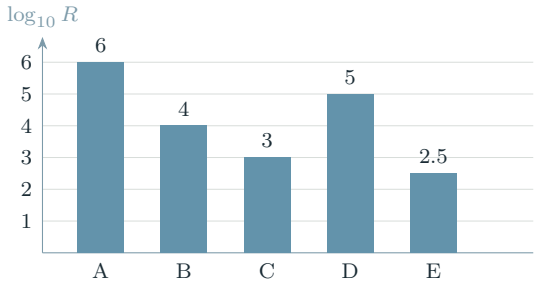
t	0	1	2	3	4
P	30	300	3000	30 000	300 000
$\log_{10} P$	1.5	2.5	3.5	4.5	5.5



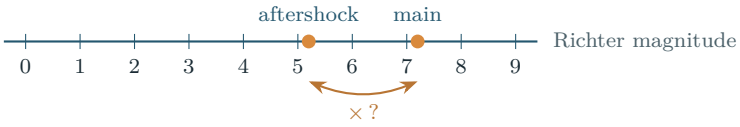
Solution. For example $\log_{10} 30 = 1.477\ldots \approx 1.5$. Each hour P is multiplied by 10, so $\log_{10} P$ goes up by exactly 1. The points lie on a **straight line**, which tells us the relationship between P and t is **exponential**.

Hurdle questions — Section 3J (show all working)

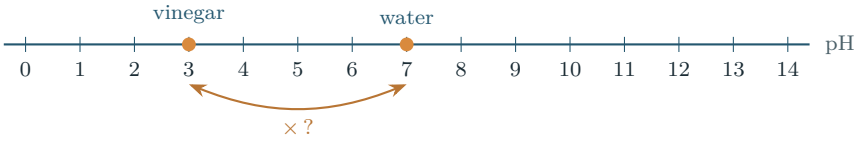
H1. The chart shows $\log_{10} R$, where R is the yearly revenue (in dollars) of five companies. How many times larger is the revenue of: **(a)** company A than company C, **(b)** company D than company B, **(c)** company B than company E? *(Part (c) is not a whole number of steps — use your calculator, to one decimal place.)*



H2. An earthquake measures 7.2 on the Richter scale and its aftershock measures 5.2. **(a)** How many times more powerful was the main earthquake? **(b)** A later tremor was 1000 times less powerful than the main earthquake. What did it measure?



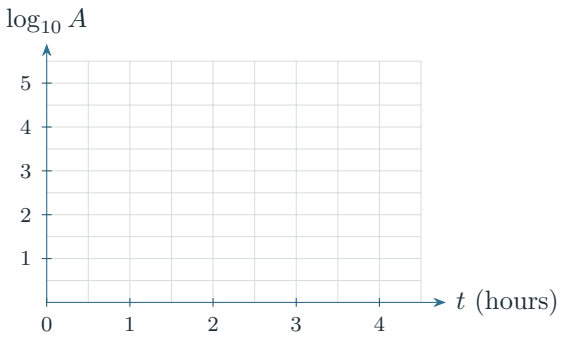
H3. On the pH scale, a *lower* pH means more acidic, and each step is a factor of 10. Vinegar has pH 3 and pure water has pH 7. **(a)** How many times more acidic is vinegar than water? **(b)** A soft drink is 100 times more acidic than water. What is its pH?



H4. State the order of magnitude of **(a)** 3 400 000 and **(b)** 0.0052. **(c)** How many orders of magnitude larger is 3 400 000 than 34, and how many times larger is that?

H5. A population of insects is counted every hour. Complete the table (one decimal place), plot $\log_{10} A$ against t on the grid, and describe what the graph tells you about the relationship between A and t.

t	0	1	2	3	4
A	200	600	1800	5400	16 200
$\log_{10} A$					



3K Laws of logarithms 10A

Textbook worked examples (cross-reference in your textbook)

- **Example 29** — Simplifying logarithmic expressions (p. 286)
- **Example 30** — Evaluating logarithmic expressions (p. 286)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 285)
- **Fluency:** Q1–4 (p. 287)
- **Problem-solving:** Q5–7 (p. 287)
- **Reasoning:** Q8–10 (p. 288)
- **Enrichment: Proving the laws for logarithms:** Q11 (p. 288)

Worked examples (study the method, then try the hurdles)

Simplifying to a single logarithm

Write each as a single logarithm: **(a)** $\log_3 5 + \log_3 4$, **(b)** $\log_2 36 - \log_2 4$, **(c)** $3\log_5 2$.

Index law	Log law	In words
$a^m \times a^n = a^{m+n}$	$\log_a x + \log_a y = \log_a(xy)$	add logs $\leftrightarrow$ multiply numbers
$a^m \div a^n = a^{m-n}$	$\log_a x - \log_a y = \log_a\left(\frac{x}{y}\right)$	subtract logs $\leftrightarrow$ divide numbers
$(a^m)^n = a^{mn}$	$\log_a x^n = n\log_a x$	a coefficient $\leftrightarrow$ a power

Solution. Use one law per line, and name it.

(a) $\log_3 5 + \log_3 4 = \log_3(5 \times 4) = \log_3 20$ (addition law)

(b) $\log_2 36 - \log_2 4 = \log_2(36 \div 4) = \log_2 9$ (subtraction law)

(c) $3\log_5 2 = \log_5 2^3 = \log_5 8$ (power law)

Combining logs, then evaluating

Evaluate without a calculator: **(a)** $\log_6 4 + \log_6 9$, **(b)** $\log_3 54 - \log_3 2$, **(c)** $2\log_{10} 5 + \log_{10} 4$.

Solution. Combine into a single log first, then ask "what power?"

(a) $\log_6 4 + \log_6 9 = \log_6 36 = 2$ $(6^2 = 36)$

(b) $\log_3 54 - \log_3 2 = \log_3 27 = 3$ $(3^3 = 27)$

(c) $2\log_{10} 5 + \log_{10} 4 = \log_{10} 25 + \log_{10} 4 = \log_{10} 100 = 2$ $(10^2 = 100)$

Hurdle questions — Section 3K (show all working)

H1. Write each as a single logarithm: **(a)** $\log_7 3 + \log_7 5$, **(b)** $\log_{10} 48 - \log_{10} 6$, **(c)** $4\log_2 3$.

H2. Evaluate without a calculator: **(a)** $\log_{10} 20 + \log_{10} 5$, **(b)** $\log_2 96 - \log_2 3$, **(c)** $\log_3 3^7$.

H3. Evaluate without a calculator: **(a)** $2\log_{10} 2 + \log_{10} 25$, **(b)** $3\log_4 2 + \log_4 8$. *(Hint: use the power law first.)*

H4. Write $\log_5 50 + \log_5 10 - \log_5 4$ as a single logarithm, then evaluate it.

H5. Sam writes $\log_{10}(20 + 80) = \log_{10} 20 + \log_{10} 80$. Show that Sam is wrong by evaluating both sides (use your calculator for the right-hand side, to three decimal places). Then write the correct law that Sam was trying to use.

3L Solving exponential equations using logarithms 10A

Textbook worked examples (cross-reference in your textbook)

- **Example 31** — Solving using the given base (p. 290)
- **Example 32** — Solving using base 10 (p. 290)

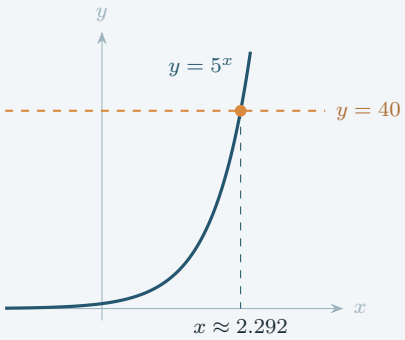
Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 290)
- **Fluency:** Q1–3 (p. 291)
- **Problem-solving:** Q4–6 (p. 291–292)
- **Reasoning:** Q7–9 (p. 292)
- **Enrichment: Change of base formula:** Q10 (p. 292)

Worked examples (study the method, then try the hurdles)

Solving $a^x = b$ with logarithms

Solve $5^x = 40$. Give the exact answer and the answer correct to three decimal places.



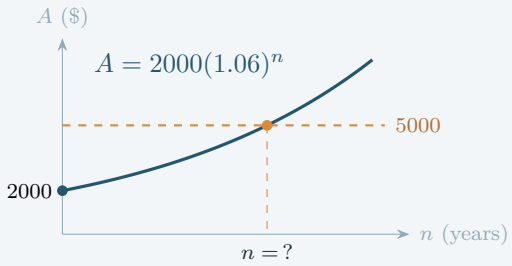
Solution. x is in the power and 40 is not a whole power of 5, so use logs. In log form, $x = \log_5 40$ (exact). With base 10 on the calculator, take $\log_{10}$ of both sides and use the power law:

$$x \log_{10} 5 = \log_{10} 40 \;\Rightarrow\; x = \frac{\log_{10} 40}{\log_{10} 5} = \frac{1.60206}{0.69897} = 2.292 \text{ (3 dp)}.$$

Check: $5^2 = 25$ and $5^3 = 125$, so x should lie between 2 and 3. ✓

Solving for time in a growth problem

\$2000 is invested at 6% p.a., compounded annually, so $A = 2000(1.06)^n$. After how many years will the investment first be worth at least \$5000?



Solution. Set up the equation, isolate the power, then take $\log_{10}$ of both sides.

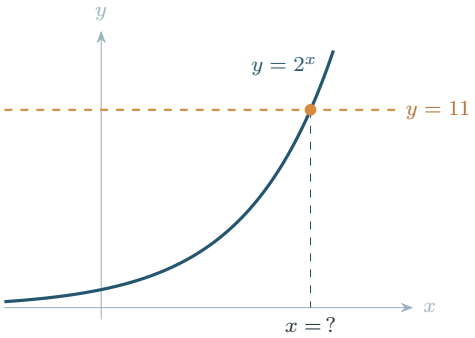
$$2000(1.06)^n = 5000 \;\Rightarrow\; 1.06^n = 2.5$$

$$n \log_{10} 1.06 = \log_{10} 2.5 \;\Rightarrow\; n = \frac{\log_{10} 2.5}{\log_{10} 1.06} = \frac{0.39794}{0.02531} = 15.73$$

Interest is added once a year, so the investment first reaches \$5000 after **16 years**.

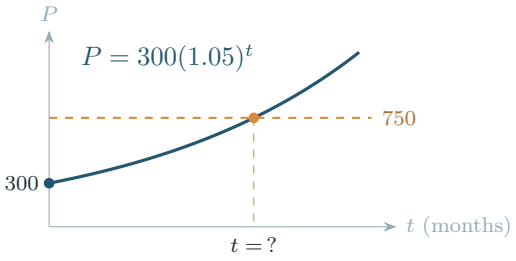
Hurdle questions — Section 3L (show all working)

H1. Solve each equation, giving the exact answer as a logarithm and the answer correct to three decimal places: **(a)** $2^x = 11$, **(b)** $3^x = 100$.

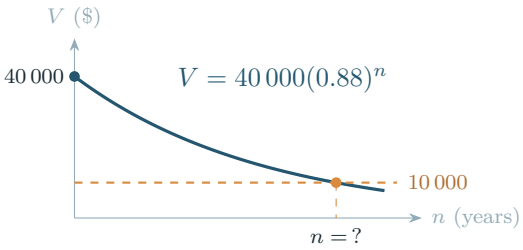


H2. Solve using base 10, correct to three decimal places: **(a)** $7^x = 300$, **(b)** $0.8^x = 0.2$.

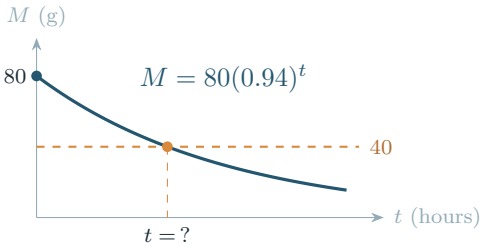
H3. A pond is stocked with 300 fish, and the population grows by 5% per month, so $P = 300(1.05)^t$. Find the number of whole months until the population first reaches 750.



H4. A car worth \$40 000 loses 12% of its value each year. After how many whole years is it first worth less than \$10 000?



H5. An 80 g sample of a radioactive substance decays by 6% every hour. Its *half-life* is the time taken for half of it (40 g) to remain. Find the half-life, correct to two decimal places.



End of Chapter 3F–3L hurdle booklet — check your working against the worked examples and ask in class if any hurdle question is unclear.